DCEG Statistical Genetics Workshop

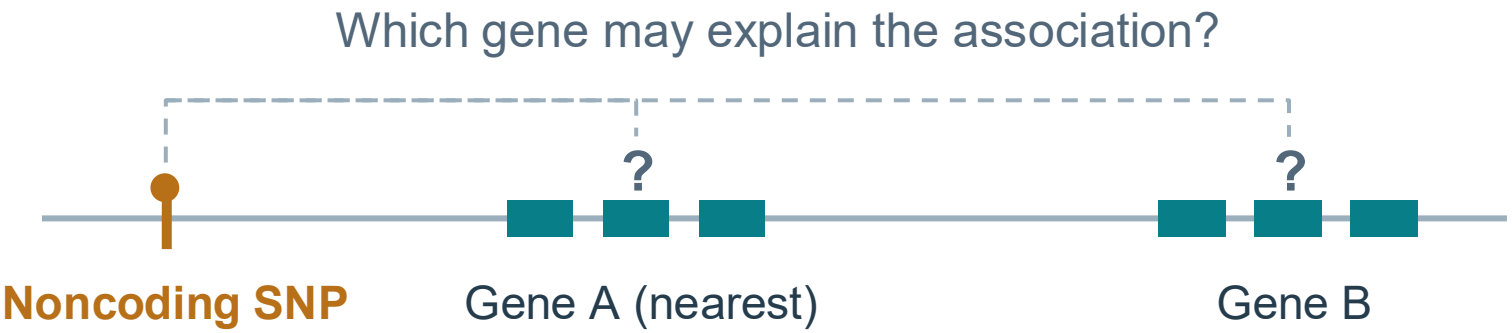
# Transcriptome-Wide Association Studies (TWAS)

From GWAS signals to gene-level hypotheses

# Why TWAS? Linking GWAS signals to genes

Most GWAS signals are noncoding, so the target gene and biological mechanism are often unclear.

**GWAS**




**TWAS**

GWAS results + expression prediction weights → Test association of predicted expression with the trait → **Candidate genes**

**No expression measurements are needed in GWAS participants.**

A TWAS association alone does not establish causality.


Hindorff et al. Proc Natl Acad Sci USA. 2009;106:9362–9367.
Gamazon et al. Nat Genet. 2015;47:1091–1098.

# The TWAS workflow

## 1 Training

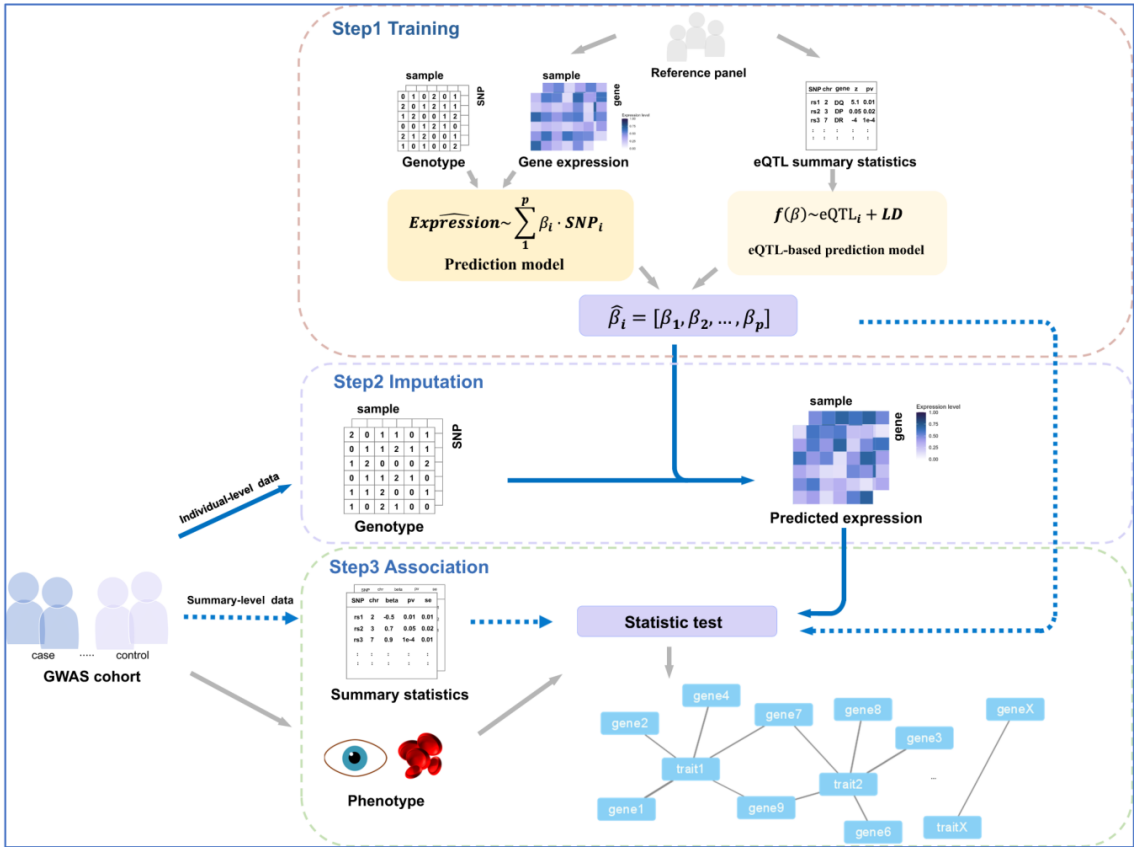
Matched genotypes and expression

## 2 Prediction

Apply SNP weights to the trait study

## 3 Testing

Associate predicted expression with the trait




Mai et al. Commun Biol. 2023;6:899. Fig. 1.

REFERENCE DATA

# GTEx: a multi-tissue expression reference

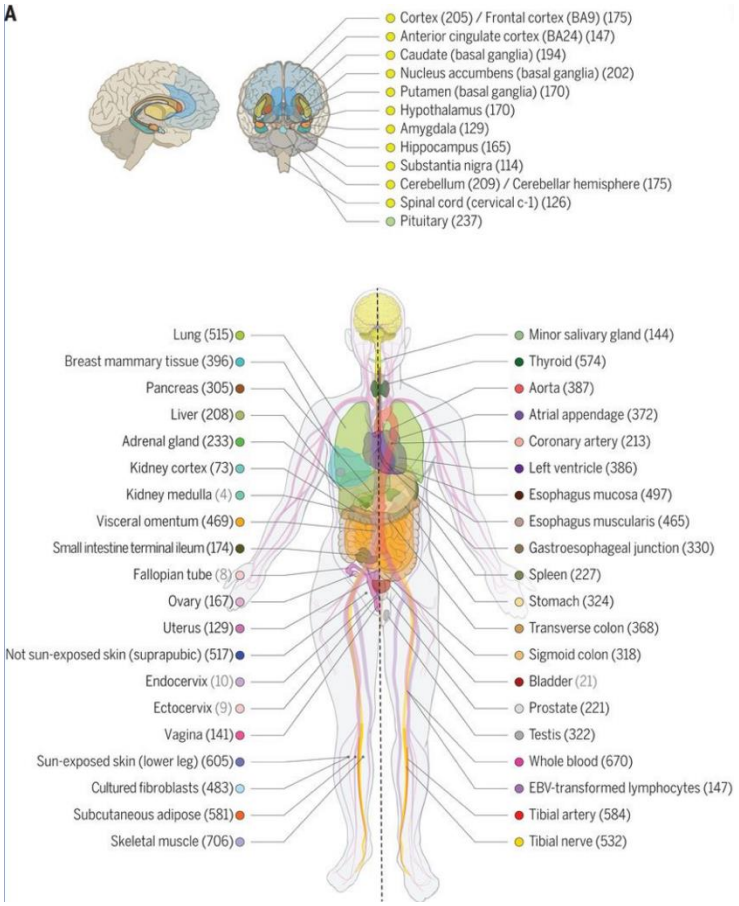
## 946 donors

GTEx v11

Matched genotype and RNA-seq data across multiple tissues

Sample size varies by tissue.

Predominantly European ancestry.




Tissue map and counts: GTEx v8

GTEx Consortium. Science. 2020;369:1318–1330. Fig. 1A.
GTEx Portal. V11 release; accessed 13 Oct 2026.

# Training gene expression prediction models

For each gene and tissue, predict expression from cis-SNPs ($\pm 500$ kb)

Taget model: $T \sim \alpha + \sum_{j=1}^{J} w_j G_j$

| Model | How SNPs enter the model |
|---|---|
| **Top SNP** | Use the strongest cis-eQTL as the only predictor. |
| **Lasso** | $L_1$ penalty selects a sparse set of SNPs. |
| **Elastic net** | $L_1$ + $L_2$ penalties can retain correlated SNPs. |

**Compare models using identical CV splits (e.g., five-fold)**

CV $R^2$ = cor²(measured expression, held-out predictions).

**One strategy: select the model with the highest CV $R^2$.**

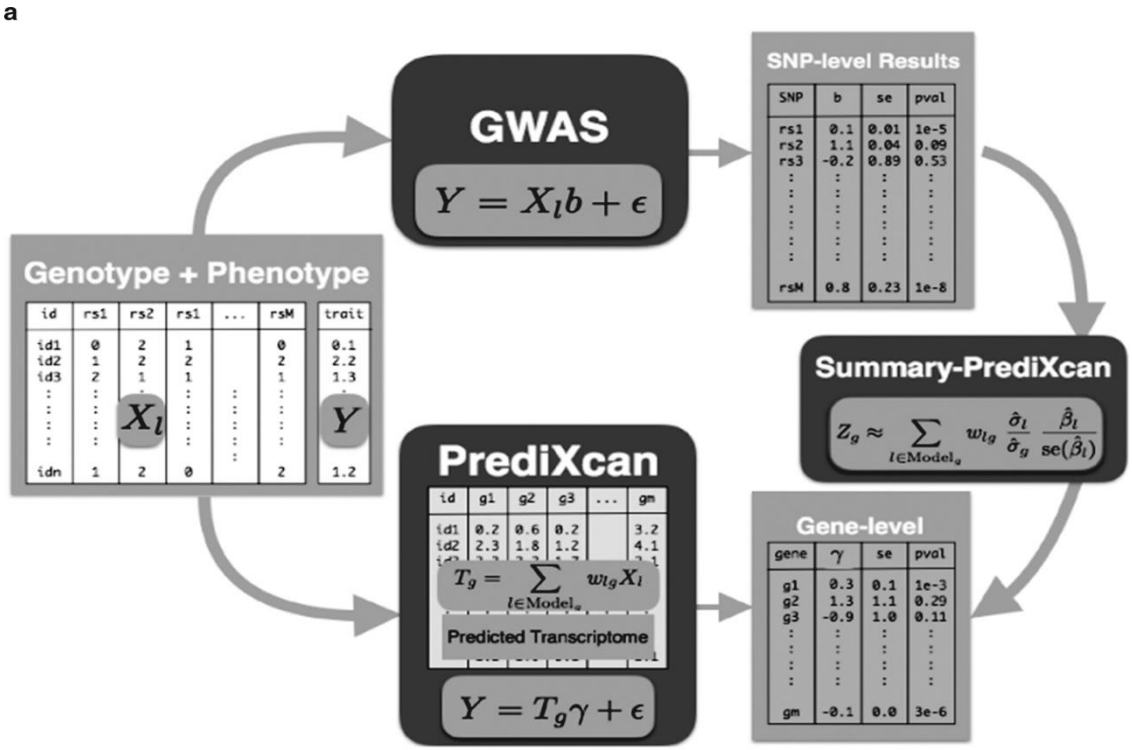
Refit on all reference samples to obtain the final SNP weights.

Select the top SNP and tune penalties within training folds.

Gamazon et al. Nat Genet. 2015;47:1091–1098.
Gusev et al. Nat Genet. 2016;48:245–252.

# Individual-level and summary-data TWAS

PrediXcan uses genotypes and traits.
S-PrediXcan uses SNP GWAS results plus reference LD.




Barbeira et al. Nat Commun. 2018;9:1825. Fig. 1A.

# The summary TWAS statistic

A weighted SNP Z-score, normalized using LD.

**Raw-dosage weights w**

$$Z_g = \frac{\sum_j w_j \, s_j z_j}{\sqrt{\mathbf{w}^\top C \mathbf{w}}}$$

C = genotype covariance

**Standardized-genotype weights**

$$Z_g = \frac{\sum_j \tilde{w}_j \, z_j}{\sqrt{\tilde{\mathbf{w}}^\top R \tilde{\mathbf{w}}}}$$

R = genotype correlation (LD)

**$\tilde{w}_j = s_j w_j$** **C = DRD** **D = diag($s_j$)**

$z_j$: SNP GWAS Z-score $s_j$: genotype standard deviation

**PrediXcan / S-PrediXcan: original genotype-dosage weights w.**
**FUSION: standardized-genotype weights $\tilde{w}$.**

Barbeira et al. Nat Commun. 2018;9:1825.
Gusev et al. Nat Genet. 2016;48:245–252.

EXAMPLE 1

# Breast cancer in African-ancestry women

## 1. Expression-model training

**150 normal-breast donors**

Komen Tissue Bank
Paired genotype and RNA-seq data

Elastic net (α = 0.5)
cis-SNPs within ±500 kb
Five-fold cross-validation

**9,982 genes with CV R > 0.10**

R = prediction correlation

## 2. Association testing

**18,034 cases**
**22,104 controls**

AABCG Consortium
GWAS summary statistics

**S-PrediXcan**

Combines trained SNP weights,
GWAS associations and LD
to test predicted expression.

**Bonferroni threshold: $P < 0.05 / 9{,}982 \approx 5 \times 10^{-6}$**

No expression measurements are needed in the GWAS participants.

Ping et al. Nat Commun. 2024;15:3718.
Barbeira et al. Nat Commun. 2018;9:1825.

EXAMPLE 1

# Two expression-TWAS associations

| Gene | Breast-cancer outcome | Z | P |
|---|---|---|---|
| TERLR1 | Overall | +4.7 | $3.2 \times 10^{-6}$ |
| EN1 | Triple-negative | −5.8 | $6.9 \times 10^{-9}$ |

Positive Z: higher predicted expression associates with higher risk.

Negative Z: higher predicted expression associates with lower risk.

Ping et al. Nat Commun. 2024;15:3718. Table 2.

INTERPRETATION

# What these associations mean

| | |
|---|---|
| **Known GWAS loci** | The gene models help prioritize candidates within associated regions. |
| **Predicted expression** | The association concerns the genetically predicted component of normal-breast expression. |
| **Follow-up analysis** | Assess colocalization and correlated gene signals, then seek functional evidence. |

Ping et al. Nat Commun. 2024;15:3718.
de Leeuw et al. PLoS Genet. 2023;19:e1010921.

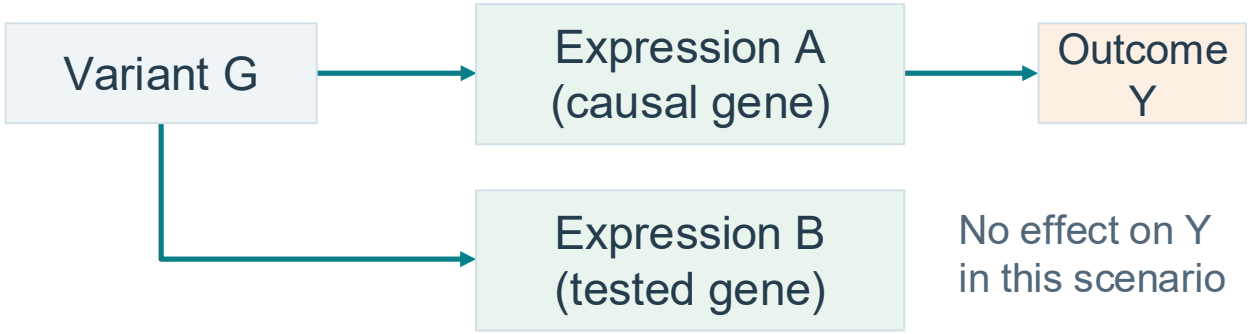
# Possible explanations for a TWAS association

## Causal mediation

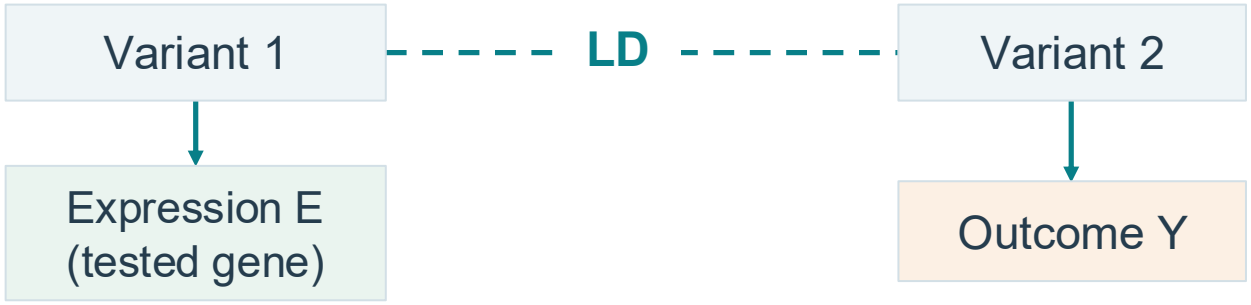



## Shared genetic regulation

Pleiotropy for gene B



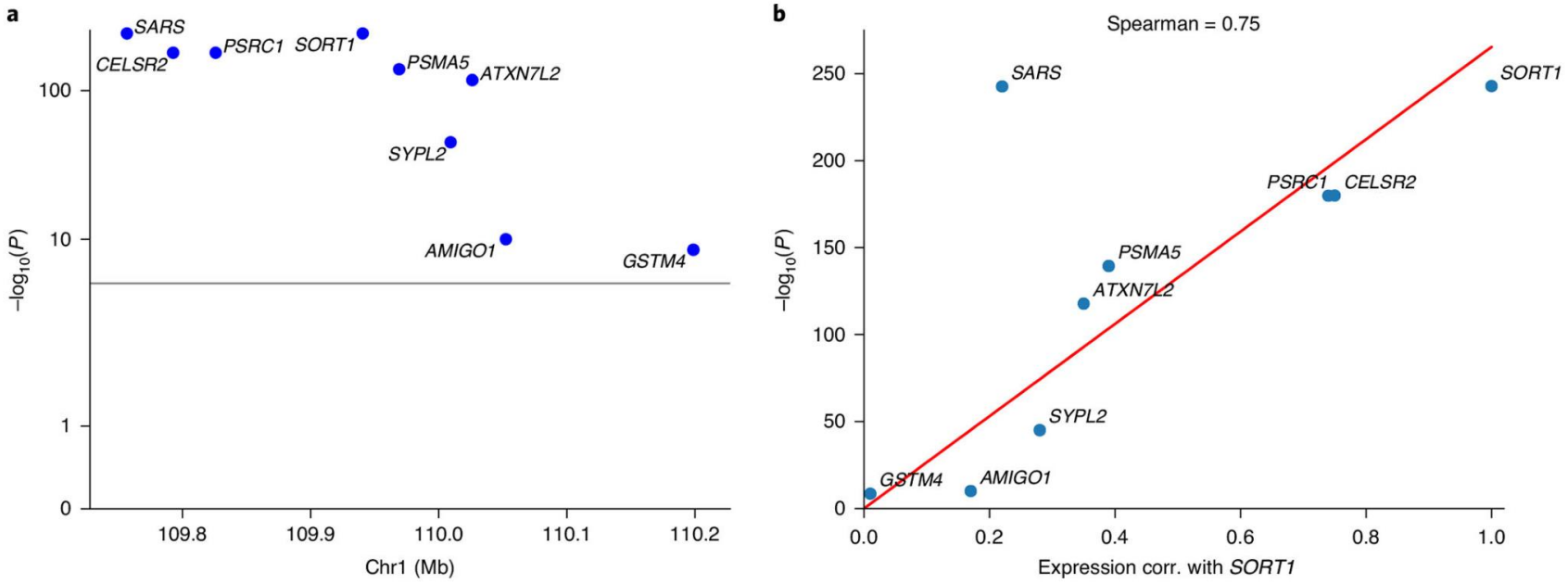

## Linkage disequilibrium




Gusev et al. Nat Genet. 2016;48:245–252.
Wainberg et al. Nat Genet. 2019;51:592–599.

# Multiple TWAS hits at the SORT1 locus

LDL cholesterol, using liver expression models




a Several neighboring genes show TWAS associations.

b TWAS evidence versus correlation with SORT1's measured expression.

**TWAS significance alone does not identify the causal gene.**

Wainberg et al. Nat Genet. 2019;51:592–599. Fig. 2.

# Where do we combine information across tissues?

| | During prediction | During association testing |
|---|---|---|
| Input | Shared and cell-specific expression components | Predicted expression across tissues |
| Model | Train expression predictors | Fit a joint trait model |
| Output | An expression predictor for each gene–cell-type pair | One omnibus P-value per gene |

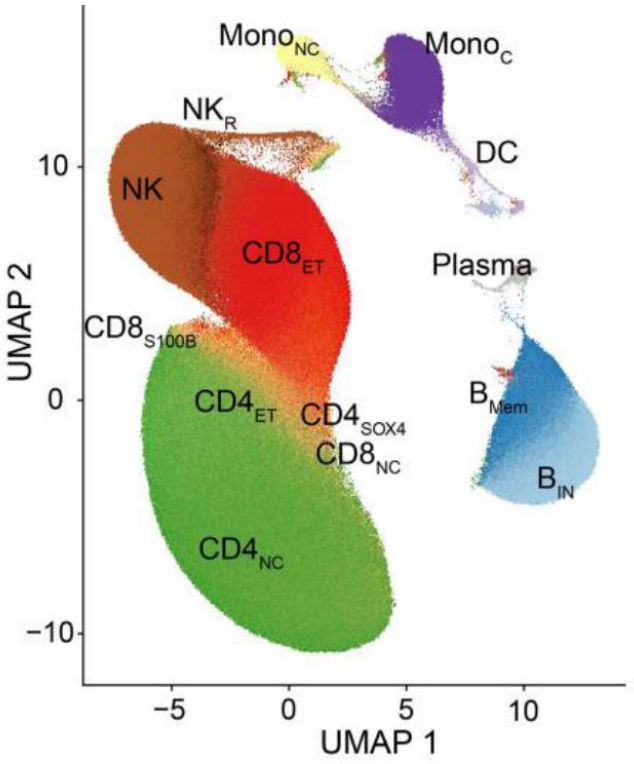
**The first improves expression prediction.**
**The second tests a gene jointly across tissues.**

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752.
Barbeira et al. PLoS Genet. 2019;15:e1007889.

EXAMPLE 2

# OneK1K: a single-cell expression reference

**~981 donors**
**~1.27 million cells**
**14 immune cell types**




Aggregate expression within each donor and cell type (pseudobulk).

**Donors are the genotype-linked training units.**

Yazar et al. Science. 2022;376:eabf3041.
Qin et al. J Natl Cancer Inst. 2026;118:1741–1752. Fig. 1A.

EXAMPLE 2

# Shared and cell-specific expression

For one gene, decompose adjusted expression within each donor.

i: donor • c: cell type • C = 14 cell types

## Shared component

$$E_i^{\mathrm{sh}} = \frac{1}{C}\sum_{c=1}^{C} E_{ic}$$

## Cell-specific component

$$E_{ic}^{\mathrm{sp}} = E_{ic} - E_i^{\mathrm{sh}}$$

**Observed expression = shared + cell-specific components**

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752. Suppl. Methods.

EXAMPLE 2

# S+allC: two-stage expression prediction

## 1 Predict the components using elastic net

**Shared component**

$$\hat{E}_i^{\mathrm{sh}} = \hat{\gamma}^{\mathrm{sh}} + \sum_{s=1}^{m} \hat{\beta}_s^{\mathrm{sh}} G_{is}$$

**Cell-specific components**

$$\hat{E}_{ic\prime}^{\mathrm{sp}} = \hat{\gamma}_{c\prime}^{\mathrm{sp}} + \sum_{s=1}^{m} \hat{\beta}_{sc\prime}^{\mathrm{sp}} G_{is}$$

m cis-SNPs with genotypes $G_{is}$ $\quad$ c′ = 1, …, C cell types

## 2 Fit target-cell expression on predicted components

$$E_{ic} = \gamma_c + w_c^{\mathrm{sh}} \hat{E}_i^{\mathrm{sh}} + \sum_{c\prime=1}^{C} w_{cc\prime}^{\mathrm{sp}} \hat{E}_{ic\prime}^{\mathrm{sp}} + \epsilon_{ic}$$

**Both stages use expression-reference data.**

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752. Suppl. Methods.

EXAMPLE 2

# Three models for each target cell type

All predict full expression. Each model fits its own coefficients.

## targetC

Direct prediction from cis-SNPs

$$\hat{E}_{ic} = \hat{\gamma}_c + \sum_{s=1}^{m} \hat{\beta}_{sc} G_{is}$$

## S+targetC

Shared + target-cell specific component

$$\hat{E}_{ic} = \hat{\gamma}_c + \hat{w}_c^{\mathrm{sh}} \hat{E}_i^{\mathrm{sh}} + \hat{w}_{cc}^{\mathrm{sp}} \hat{E}_{ic}^{\mathrm{sp}}$$

## S+allC

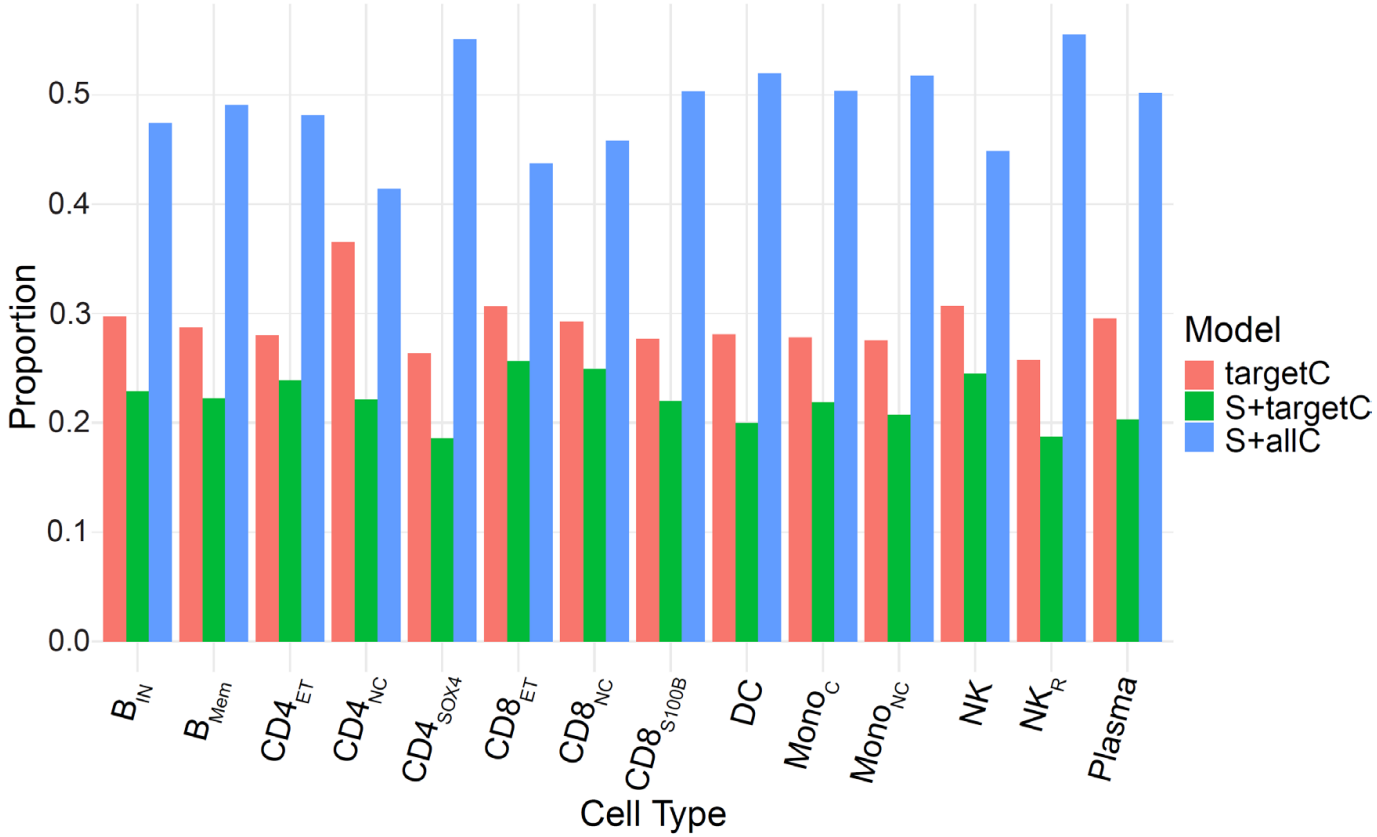
Shared + specific components from all cell types

$$\hat{E}_{ic} = \hat{\gamma}_c + \hat{w}_c^{\mathrm{sh}} \hat{E}_i^{\mathrm{sh}} + \sum_{c'=1}^{C} \hat{w}_{cc'}^{\mathrm{sp}} \hat{E}_{ic'}^{\mathrm{sp}}$$

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752.
Thompson et al. Nat Commun. 2022;13:5704.

EXAMPLE 2

# S+allC most often predicts expression best

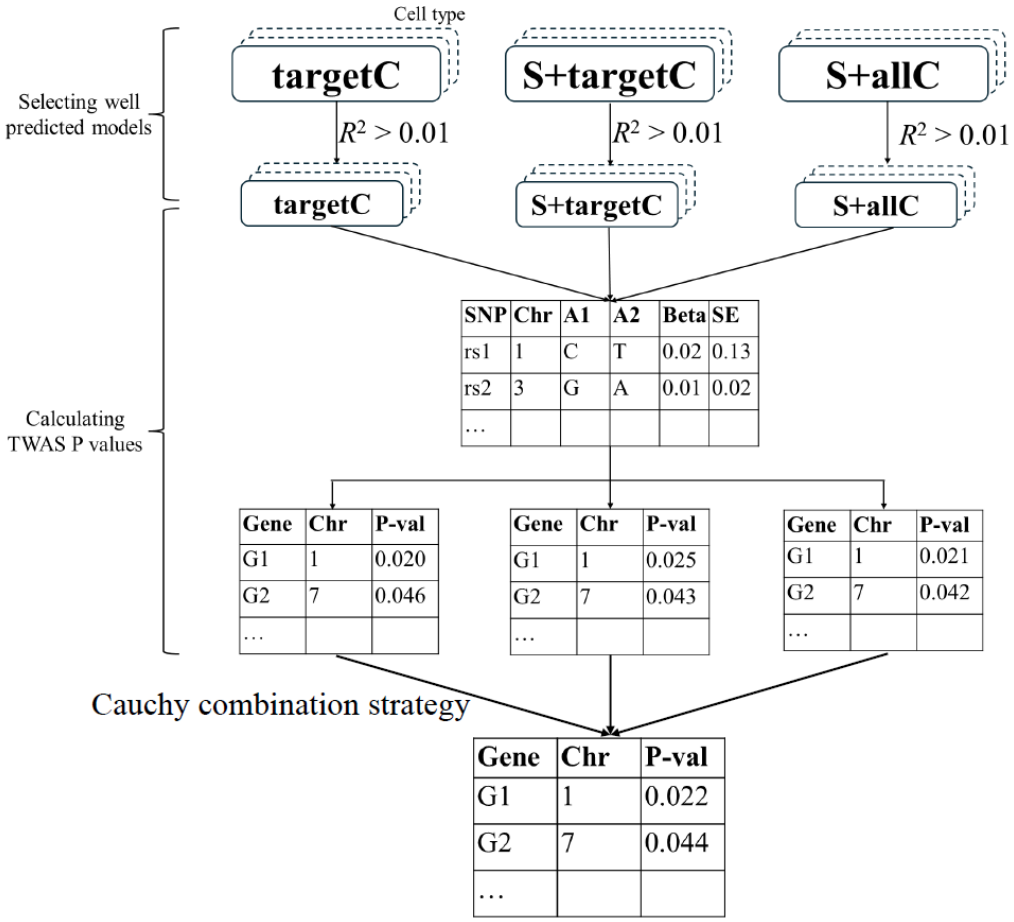



Proportion of genes for which each model has the highest CV $R^2$.

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752. Fig. S3.

EXAMPLE 2

# Combining TWAS evidence across prediction models




## Retain eligible models

CV $R^2 > 0.01$ (10 folds)
Keep every eligible model.

## Calculate TWAS P-values

## Cauchy combination

One combined P-value for each gene–cell-type pair.

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752. Fig. S2.

EXAMPLE 2

# Breast cancer: datasets and sample sizes

| Purpose | Dataset | Data used |
|---|---|---|
| Expression prediction | OneK1K | 981 donors<br>~1.27 million cells<br>14 immune cell types |
| Discovery | BCAC | 118,474 cases<br>96,201 controls |
| External follow-up | UK Biobank | 7,968 cases<br>186,185 controls |

**LD reference: 1000 Genomes European panel (hg19)**

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752. Table S1.

EXAMPLE 2

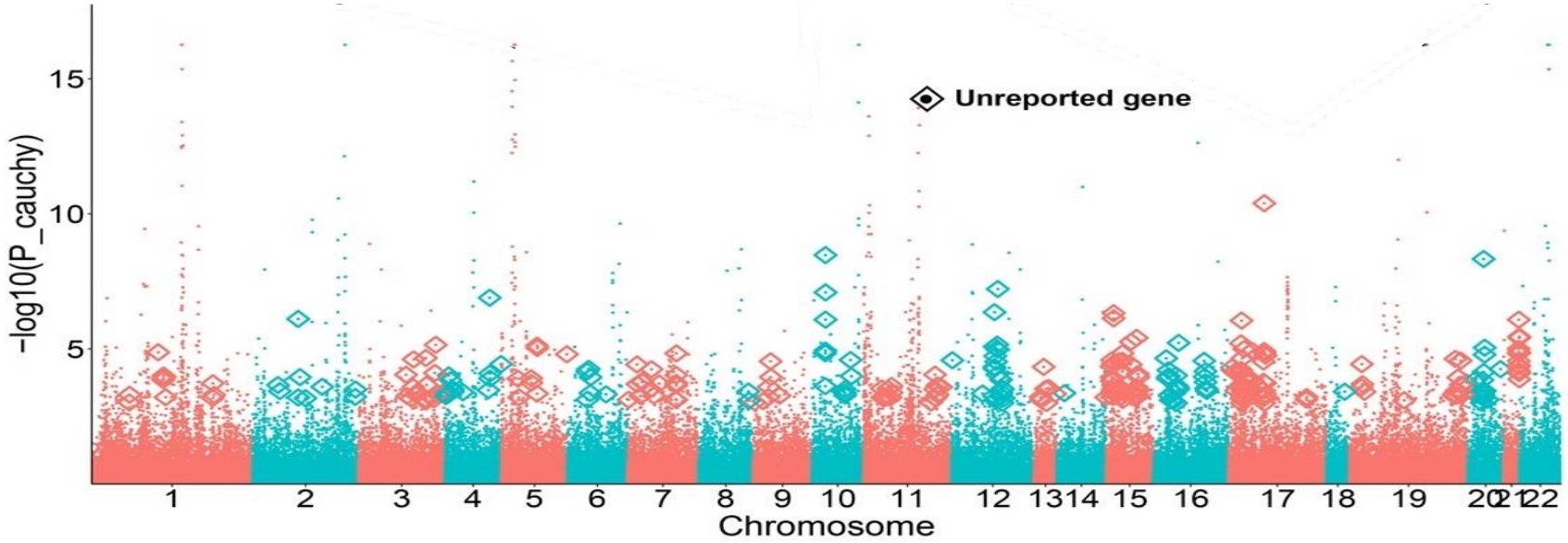
# Breast-cancer signals across immune cells

## 240 genes

Bonferroni significant

## 16 genes

GWAS-unreported by prior GWAS




Results across 14 immune cell types. Counts refer to unique genes.

Qin et al. J Natl Cancer Inst. 2026;118:1741–1752. Fig. 2B; Table S4.

EXAMPLE 3

# MultiXcan: the high-cholesterol example

## Expression-model training

**GTEx v6p: 44 tissues**

Separate elastic-net model for each gene–tissue pair

Mixing parameter α = 0.5

10-fold CV selects penalty λ

Original v6p filter: q < 0.05

## Application in UK Biobank

Self-reported high cholesterol

**50,497 cases**

**100,994 controls**

Predict expression from genotypes in up to 44 tissues per gene.

**17,434 genes tested**

## PCA based joint test: linear regression on retained PCs

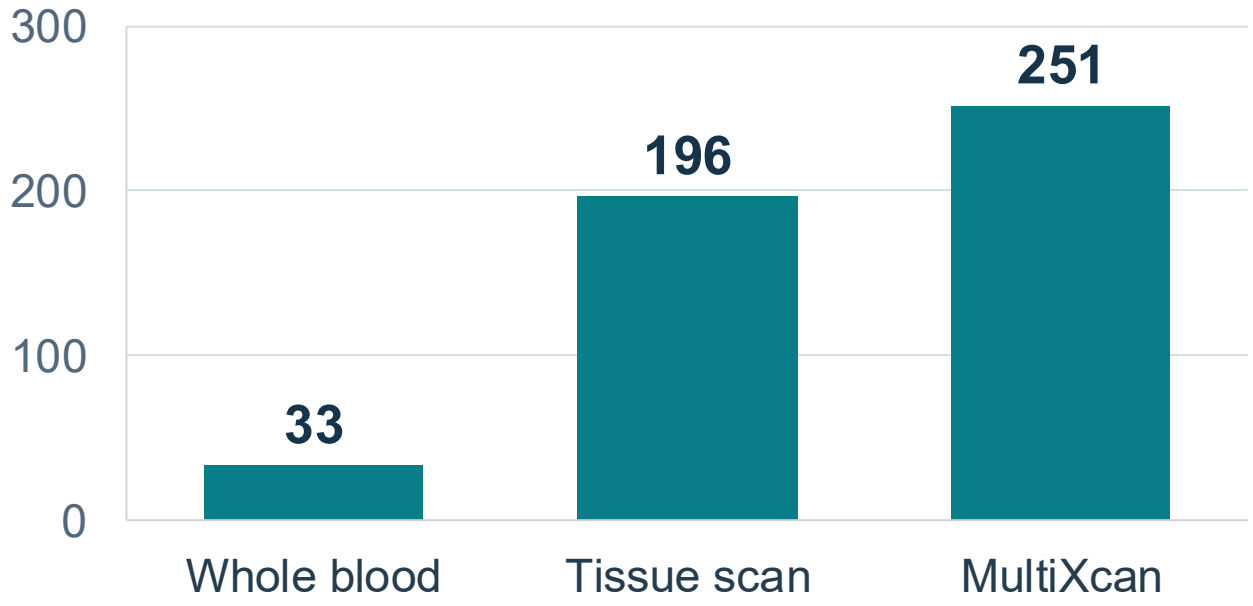
$$Y \sim \alpha + \sum_{k=1}^{44} \beta_k \hat{E}_k$$

Barbeira et al. PLoS Genet. 2019;15:e1007889.
Barbeira et al. Nat Commun. 2018;9:1825.

EXAMPLE 3

# MultiXcan detected 251 associated genes

Bonferroni-significant genes in the high-cholesterol analysis




**+55**

net additional genes

28% more than the tissue scan

Gene coverage and Bonferroni thresholds differ across approaches.

Barbeira et al. PLoS Genet. 2019;15:e1007889. Fig. 2.

# TWAS limitations and follow-up analyses

| Limitation | Why it matters | What to do next |
|---|---|---|
| **Prediction and coverage** | Weak models miss signals or leave genes untested. | Check CV $R^2$ and which genes have usable prediction models. |
| **Tissue or cell type** | Shared regulation can yield signals in several tissues. | Compare biologically relevant tissues and cell types. |
| **Ancestry and LD** | Prediction and LD patterns can differ across populations. | Check model portability and use a suitable LD reference. |
| **Correlated gene signals** | Shared eQTLs or LD can make genetic predictors hard to distinguish. | Conditional TWAS, gene-level fine-mapping, MR. |

Porcu et al. Nat Commun. 2019;10:3300.
Liu et al. Nat Genet. 2024;56:348–356.

TAKE-HOME MESSAGES

# What to remember

**The test** — TWAS associates genetically predicted expression with a trait.

**The reference** — Prediction quality, tissue context, ancestry, and LD affect what can be detected.

**The extension** — Combining tissues during prediction and during association answers different questions.

**A significant gene is a candidate for causal follow-up.**

Gusev et al. Nat Genet. 2016;48:245–252.
de Leeuw et al. PLoS Genet. 2023;19:e1010921.